

HDC Python Golden Reference

Bit-exact reference model for the 1024HDC RTL (`xor_permute_top` , `permute_stage` , future `bundle_unit` , `popcount_am` , `pruning_mask`).

Setup

```
cd C:\Modelsim_projects\1024HDC\python_ref
python -m pip install -r requirements.txt
```

Quick smoke test

```
python run_smoke_test.py
```

Generate ModelSim stimulus / expected outputs

```
python generate_vectors.py --out-dir vectors --count 1000 --seed 42
```

Library usage

```
from hdc_ref import HDCConfig, HDCEngine, ItemMemory, train_class_hypervectors

cfg = HDCConfig(D=1024, seed=42)
engine = HDCEngine(cfg)
mem = ItemMemory(cfg)

# bind + permute (matches xor_permute_top)
out = engine.bind_permute(in_vec, bind_vec, perm_mode=2, perm_param=73)

# EMG-style record encoding
query = engine.encode_emg_window(quantized_features, mem)

# Twist 1: informed vs random pruning masks
informed, random = engine.make_pruning_masks(query_hvs, labels, keep_ratio=0.5,
seed=0)
```

Files

File	Purpose
<code>hdc_ref.py</code>	Core library

File	Purpose
<code>generate_vectors.py</code>	Export hex vectors for co-simulation
<code>run_smoke_test.py</code>	Local regression without pytest
<code>vectors/</code>	Generated co-sim files (created on demand)